# BiasAperture — Demographic Fairness Audit Dossier

Diagnostic Demographic Bias Evaluation & Regulatory Traceability Report

AUDITED MODEL
**HuggingFace ViT-B/16 Gender Classifier**

EVALUATION DATASET
**FairFace Benchmark Dataset (N=10954)**

PROTECTED AXIS
**Race**

AUDIT DATE
**2026-10-18 16:46:53 UTC**

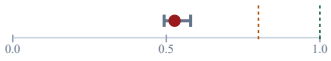
**Diagnostic Scope & Non-Intervention Guarantee:**

BiasAperture is engineered exclusively as an external diagnostic auditing, statistical significance testing, and regulatory traceability platform. It does **not** perform model retraining, in-processing debiasing, or synthetic image generation. In adherence to statistical integrity safeguards (NFR-003), demographic subgroups with fewer than 30 samples ($n < 30$) have point estimates suppressed to prevent unreliable compliance claims.

| 7 / 7 | Overall (Table 1) | DIR = 0.527 | 0 |
|---|---|---|---|
| SUBGROUPS WITH SIGNIFICANT DISPARITY (P < 0.05) | WORST DISPARITY COHORT (28 METRIC FAILURES) | WORST HEADLINE METRIC (OVERALL) | SUBGROUPS FLAGGED N < 30 (NFR-003) |

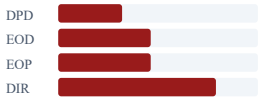
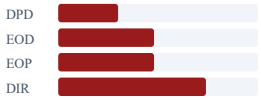
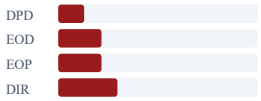
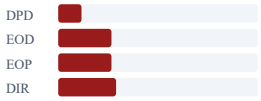
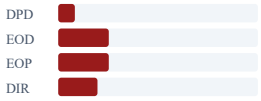
## 1. Headline Fairness Metrics (Core Four — Cross-Group Overview)

Summary metrics evaluated across all eligible demographic strata (n ≥ 30 per NFR-003). Accompanied by 95% stratified BCa bootstrap confidence intervals and Pearson's $\chi^2$ test of independence. Plotted on a calibrated [0.0, 1.0] domain with null-point reference and threshold lines.

| Metric Name | Point Estimate | 95% Bootstrap CI | CI Band vs. Null | Significance (p-value) | Regulatory Clause |
|---|---|---|---|---|---|
| **Demographic Parity Difference** | 0.276 | [0.239, 0.308] |  | p=0.0000 **p < 0.05** | EU AI Act Art. 10(2)(f) / NIST Measure 2.11 |
| **Equalized Odds Difference** | 0.274 | [0.232, 0.305] |  | p=0.0000 **p < 0.05** | EU AI Act Art. 10(2)(g) / NIST Measure 2.11 |
| **Equal Opportunity Difference** | 0.274 | [0.232, 0.305] |  | p=0.0000 **p < 0.05** | EU AI Act Art. 10(3) / NIST Measure 2.11 |
| **Disparate Impact Ratio** | 0.527 | [0.493, 0.579] |  | p=0.0000 **p < 0.05** | EEOC 4/5ths Rule / EU AI Act Annex IV §2(g) |

## 2. Per-Subgroup Disparity Matrix

Detailed stratification across individual demographic groups, ordered by largest deviation from the null (no-disparity) point, most significant first. Subgroups below the NFR-003 sample threshold (n < 30) are strictly flagged, uncomputed to prevent small-sample estimation artifacts, and summarised separately below.

| Subgroup | Sample Size (n) | DPD | EOD (Max-Gap) | EOP | DIR (Symmetric) | Deviation from Null (red = p < 0.05) | Data Integrity Status |
|---|---|---|---|---|---|---|---|
| **Middle Eastern** | 1209 | 0.181 | 0.131 | 0.131 | 0.691 |  | Audited (Valid) |
| **East Asian** | 1550 | 0.096 | 0.139 | 0.139 | 0.763 |  | Audited (Valid) |
| **Southeast Asian** | 1415 | 0.090 | 0.144 | 0.144 | 0.778 |  | Audited (Valid) |
| **White** | 2085 | 0.039 | 0.065 | 0.065 | 0.911 |  | Audited (Valid) |
| **Black** | 1556 | 0.035 | 0.080 | 0.080 | 0.913 |  | Audited (Valid) |
| **Indian** | 1516 | 0.025 | 0.076 | 0.076 | 0.941 |  | Audited (Valid) |
| **Latino_Hispanic** | 1623 | 0.005 | 0.050 | 0.050 | 0.988 |  | Audited (Valid) |

## 3. Governance Documentation & Intended Use

> **EU AI Act (Regulation EU 2024/1689) Article 10 / Annex IV Alignment:** This audit dossier provides verifiable diagnostic documentation of demographic performance disparities. It does not perform model retraining or in-situ weight modifications.

**Model Description:** Vision Transformer (ViT-B/16) pre-trained on ImageNet-21k / ImageNet-1k and fine-tuned on FairFace, evaluating self-attention mechanisms against demographic disparity propagation.

**Backend configuration:** dual. Both backend executions completed.

Cross-backend agreement checks point estimates only; bootstrap, hypothesis tests, and reporting share implementation. Consult execution logs for backend failures and divergences.